

Demo: Chewpin: a wearable acoustic device for chewing detection

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ABSTRACT

Diet intervention has emerged as a promising strategy in obesity prevention and treatment. Existing research predominantly focused on macronutrient intake and food quantity restriction in diet manipulation. Eating habit is difficult to change due to its highly habitual nature; therefore, it is essential to automatically detect eating behavior and provide real-time intervention in unhealthy eating patterns. In this study, we explored the possibility of designing Chewpin, an easy-to-be-implemented and socially acceptable device for capturing eating behavior (i.e., chewing and swallowing) in a controlled environment. We implemented a convolutional neural network (CNN) for data classification. Overall, our system achieved a promising accuracy of eating recognition of 98.23% on the test set. In the future, we will evaluate its usability and feasibility in real-life eating practices and use this system as a technical tool for problematic eating intervention.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

eating detection, acoustic sensing, wearable device, CNN

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1 INTRODUCTION

Overweight and obesity have become one of the most significant public health issues in societies [4]. Accumulating evidence suggested that macronutrient intake is one of the leading causes of unhealthy weight gain [3]. Thus, researchers strived to monitor eating activity as a means to gather information on problematic eating and implement dietary intervention to help people regularize eating

habits and take back control of their health. One major challenge for eating intervention is to understand when people eat. Researchers in the area of eating event detection have proposed several techniques to determine eating-related behaviors. For instance, audio sensors were used for collecting acoustic information of chewing and swallowing in the ear canal or on the throat [1, 2, 6], camera for first-person image analysis of eating behavior [5, 7, 9], wrist-based eating gesture recognition [8, 10]. Although these systems have been widely explored in eating detection, they have several practical limitations such as obtrusive, privacy-invasive, and not social acceptability in real-life eating practice. In this preliminary study, by weighing the pros and cons, we explored the usability of acoustic sensing, an easy to be implemented and socially-acceptable technique for eating detection. Two major design considerations should be met: 1. It should be suitable to be worn in the real world: comfortable and socially acceptable. 2. It should capture information-rich features from audio and accurately distinguish eating episodes in a given scenario. With these concerns in mind, we designed a wearable acoustic device, Chewpin, as a technical tool for audio data collection. Then, we implemented this device in the eating sound collection in a lab-controlled environment and extracted features to train a convolutional neural network (CNN) to classify eating and non-eating behavior. To the best of our knowledge, no study has applied CNN for acoustic eating sound detection before. The accuracy of our system showed a promising result of 98.23% on the test set. The contribution of this ongoing exploratory study are:

- Design and implement a wearable acoustic sensor-based device that can capture eating behavior (i.e., chewing, swallowing).
- Develop and evaluate an eating detection classifier model using CNN based on a set of acoustic features.

2 SYSTEM DESIGN

2.1 Hardware device

Acoustic microphone A dual-microphone expansion board (ReSpeaker 2-Mics Pi HAT) was applied to capture eating sound. The board is developed based on WM8960, low-power stereo codes, and two microphones which are capable of collecting nuanced acoustic sounds supporting a max 48kHz sampling. This device has been used in various AI assistant and voice interaction applications. **Mini computer** Raspberry Pi Zero W was chosen as a tiny, low-cost mini-computer with on-board wireless LAN and Bluetooth 4.1 for hardware prototyping. This mini-computer is compatible with a dual-microphone and is feasible to be implemented in real-living conditions as a wearable device for eating data collection (Figure 1).

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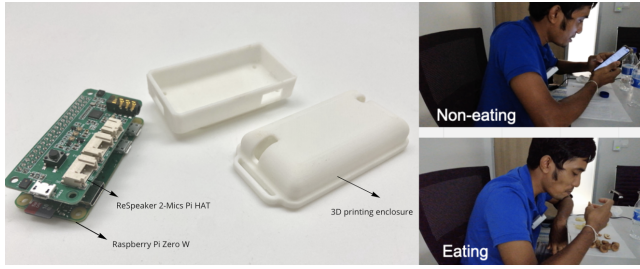


Figure 1: Hardware components of Chewpin

2.2 Data collection

Procedure. 8 participants (3 female), age (26.75 ± 2.1), BMI (21.75 ± 2.7) were recruited for data collection. Before the experiment, participants were asked to fill out the consent form and basic demographic information (e.g., age, gender, BMI). Five test food were specifically chosen that were commonly consumed in daily eating practice, see Figure 2. Participants were guided to wear the device on their collar in a private room and perform designated activities.

Data annotation and ground truth. In order to simplify the data annotation process, we designated a series of eating and non-eating activities that participants were asked to perform in a randomized sequence, see Figure 2. To know when participants actually ate, we applied video recording for ground-truth measurement.

Activity	Description	Number of segments	Total time length
Eating	Eating serval sushi	2	2 min
	Eating serval photo chips	2	2 min
	Eating serval nuts	2	2 min
	Eating serval biscuits	2	2 min
	Take a moment to relax	1	2 min
Silence	Take a moment to relax	1	2 min
Drinking	Drinking water	8	4 min
Talking	Read an article	1	2 min

Figure 2: Test food and activity list in the experiment

2.3 Audio classification

Historically, researchers on audio detection with features such as spectrograms, MFCCs and used classifiers based on HMMs, SVMs. In our study, we firstly applied CNN for eating detection by interpreting audio as an image.

Pre-processing For data extraction, mel scale spectrograms were applied, see Figure 3. The core idea of this logarithmic transformation is to mimic how humans perceived sound. 2500 samples(3 seconds) from 144 audio segments were extracted. Hanning window and short-time Fourier transformation were applied to decompose the signal into the frequency domain. Librosa were used to convert data to mel scale. We applied z-score normalization and the min-max scaling to decompose data values between 0 and 255 as image frames for classifier training.

Model training Mel spectrograms were used as an input for the deep learning model. Specifically, we used Resnet 34 as a pre-trained model. In our case, the third dimension is set as 1. CrossEntropy and Adam were used for the training model. Finally, the accuracy of the model on the test set was 98.23%.

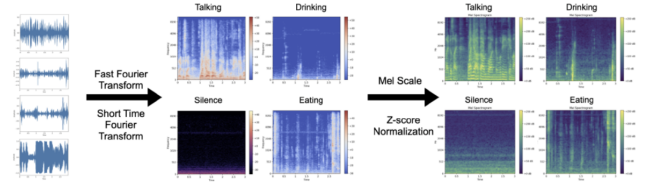


Figure 3: Pipeline of data pre-processing and feature extraction

3 LIMITATION AND FUTURE WORK

There are several limitations to our study. For instance, given that an acoustic microphone was chosen for this exploration, data collection was restricted in controlled lab settings; thus, there is a significant gap with real-life eating situations. Although the model has a promising test accuracy on the test set, how the model will detect and classify real-time eating behavior was not addressed.

In the future, we will add free-living eating data and explore the combination of other modalities such as inertial sensing for gesture recognition to achieve both high accuracy and wearability of eating detection in real-life eating practice. Moreover, we will continue to explore the potential for implementing this eating detection system as a wearable technique for problematic eating intervention.

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